

Bedrock Programmer's Reference Manual

SAVE/RESTORE Processor-State Save Area Preview

6 SAVE/RESTORE PROCESSOR-STATE SAVE AREA

Save-area bytes								Offset		
63..56	55..48		47..40	39..32	31..24	23..16	15..8	7..0		
reserved		DBANK	STATUS		FLAGS		reserved		GS_VALID	0x000
state block bitmap										0x008
D0, bank 0										0x010
D1, bank 0										0x018
D2, bank 0										0x020
D3, bank 0										0x028
D4, bank 0										0x030
D5, bank 0										0x038
D6, bank 0										0x040
D7, bank 0										0x048
A0										0x050
A1										0x058
A2										0x060
A3										0x068
A4										0x070
A5										0x078
A6										0x080
A7										0x088
GS0										0x090
GS1										0x098
GS2										0x0a0
GS3										0x0a8
GS4										0x0b0
reserved padding to end of fixed base block										0x0b8
extension state block area may begin										0x0c0

Base Header Fields:

Offset	Bits	Field	Meaning
0x000	0..4	GS_VALID	bit n validates GS _n at 0x090 + 8*n; RESTORE updates only GS slots whose bits are set
0x000	5..15	reserved	SAVE writes zero and RESTORE ignores
0x000	16..31	FLAGS	always-saved integer condition-code FLAGS image
0x000	32..47	STATUS	always-saved STATUS image; user-mode RESTORE ignores this field, supervisor RESTORE applies STATUS write rules
0x000	48..51	DBANK	current data-register bank selector; RESTORE applies the SELDB validity rules and updates the selector after bank payloads
0x000	52..63	reserved	SAVE writes zero and RESTORE ignores
0x008	i	component descriptor 2+i present	i=0..SAVE_COMPONENT_COUNT-1 and SAVE_COMPONENT_COUNT must be no larger than 64
0x008	all other bits	reserved	SAVE writes zero and RESTORE ignores

SAVE always writes the base header fields, state block bitmap, bank-0 D registers, A registers, FLAGS, STATUS, and DBANK selector. SAVE may clear GS validity bits for architecturally clean GS slots and may clear state-block bitmap bits for architecturally clean extension blocks. RESTORE always restores bank-0 D registers, A registers, FLAGS, and DBANK from the fixed base block. RESTORE updates only GS slots whose GS validity bit is set. User-mode RESTORE ignores the STATUS field and does not validate STATUS reserved bits; supervisor RESTORE restores STATUS under the normal STATUS write rules. RESTORE processes extension blocks whose state-block bitmap bit is set and ignores blocks whose bit is clear.

Extension component offsets begin no earlier than SAVE_HEADER_SIZE, 0x0c0 for the base profile, and are reported by CPUID. The save-area base address is 4 KiB aligned. Every component offset and component storage size is a multiple of 64 bytes so state blocks are cache-line aligned.

CPUID SAVE_AREA_LAYOUT component descriptors are indexed from 2 upward. That descriptor index order is the architectural extension-component order and is the bit order used by the fixed state block bitmap at offset 0x008. COMPONENT_OFFSET values are numeric byte offsets and must be strictly increasing in descriptor order. Base-defined component formats appear first; extension-owned component formats are appended by descriptor order from their owning extension specs. The SAVE area has no architectural total-size upper bound beyond the CPUID-reported SAVE_AREA_SIZE value for the selected implementation/profile, but SAVE_COMPONENT_COUNT is limited to 64 by the fixed state block bitmap.

Extension Component Formats:

Extension Component: Additional Data-Register Banks

Component ID 0x0001
Component Name DATA_REGISTER_BANKS
Component Size 0x40 * DBANK_COUNT

This component stores data-register banks other than bank 0. Bank 0 is part of the fixed base block. The first 64 bytes are the component header. Each following 0x40-byte bank record stores D0-D7 for the matching bank number and begins on a 64-byte boundary.

The component header fields contain a bank-level validity bitmap. If a bank bit is clear, RESTORE leaves that nonzero D-register bank unchanged.

Component-relative bytes								Offset
63..56	55..48	47..40	39..32	31..24	23..16	15..8	7..0	
reserved						D_BANK_VALID		+0x000
reserved padding to first bank record								+0x008
D bank records								+0x040

Component Header Fields:

Offset	Bits	Field	Meaning
+0x000	0	reserved	bank 0 lives in the fixed base block
+0x000	1..15	D_BANK_VALID	bit b validates D bank b at +0x040*b for b=1..DBANK_COUNT-1
+0x000	16..63	reserved	bits for unimplemented banks and future state are reserved

Extension Component: Floating-Point State

Component ID 0x0002
Component Name FLOATING_POINT
Extension Requirement floating_point
Component Size 0x00c0

This component stores the floating-point accrued exception flags, the floating-point status/control-mode word, and the architectural floating-point register file. The first 64-bit word packs the F-register validity bitmap together with FFLAGS and FSTATUS. The F register file starts at +0x008 and the component storage size is rounded up to a 64-byte boundary.

The component header contains the F-register validity bitmap. FFLAGS and FSTATUS are present whenever the floating-point state block is present.

Component-relative bytes								Offset
63..56	55..48	47..40	39..32	31..24	23..16	15..8	7..0	
FSTATUS		FFLAGS		reserved		F_VALID		+0x000
F0								+0x008
F1								+0x010
F2								+0x018
F3								+0x020
F4								+0x028
F5								+0x030
F6								+0x038
F7								+0x040
F8								+0x048
F9								+0x050
F10								+0x058
F11								+0x060
F12								+0x068
F13								+0x070
F14								+0x078
F15								+0x080
reserved padding to 0x0c0								+0x088

Component Header Fields:

Offset	Bits	Field	Meaning
+0x000	0..15	F_VALID	bit n validates Fn at +0x008 + 8*n
+0x000	16..31	reserved	SAVE writes zero and RESTORE ignores
+0x000	32..47	FFLAGS	floating-point accrued exception flags, always present when this state block is present
+0x000	48..63	FSTATUS	floating-point control and status-mode register, always present when this state block is present